



International University, VNU-HCMC

School of Computer Science and Engineering

Lecture 5 Statistical and Machine Learning Techniques for Big Data (part 2)

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Purpose of the lecture

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- Extend foundational machine learning knowledge to advanced techniques suitable for Big Data.
- Introduce advanced supervised and unsupervised learning methods for complex, large-scale datasets.
- Explain the role of dimensionality reduction and feature engineering in improving model performance and scalability.
- Connect machine learning techniques with real-world Big Data applications and case studies.



Warm-up question

1. In Part 1, we learned basic supervised and unsupervised learning methods. What limitations might these methods face when applied to very large or high-dimensional datasets?
2. If a dataset contains millions of records and thousands of features, which challenges do you expect when building machine learning models, and why might more advanced techniques be needed?



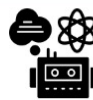
Outlines



Recap of Part 1 &
Motivation



Advanced
Supervised
Learning



Advanced
Unsupervised
Learning



Dimensionality
Reduction



Feature
Engineering &
Feature Selection



Machine Learning
at Scale



Applications &
Case Studies



Recap of Part 1 & Motivation

Statistical foundations for Big Data



Descriptive statistics



Probability & distributions



Sampling and basic inference

6



Descriptive Statistics



Summarize and describe
key characteristics of
data.



Common measures: mean,
median, variance,
standard deviation.



Help understand data
distribution and detect
anomalies.

7



Probability & Distributions



Model uncertainty and randomness in data.



Common distributions: normal, binomial, Poisson.



Provide the basis for prediction and decision-making.



Sampling and Basic Inference



Use samples to conclude large populations.



Reduce computation cost in Big Data analysis.



Support estimation and hypothesis testing.



Limitations of Basic Techniques

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- Assume simple data structures (linear boundaries, spherical clusters)
- Performance degrades with:
 - High dimensionality
 - Large-scale datasets
 - Noisy and complex data
- Limited scalability and robustness in real-world Big Data systems

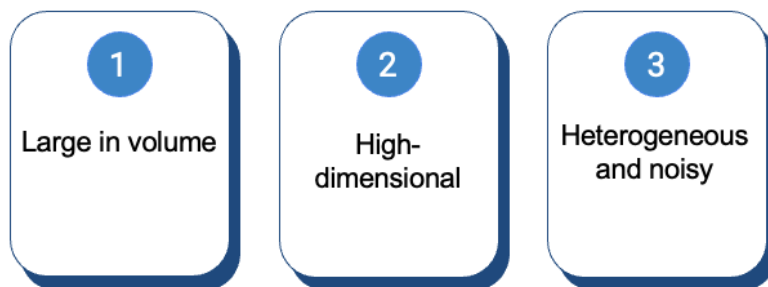
10



Motivation for Advanced Techniques

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Big Data is often:



11



Motivation for Advanced Techniques

Need methods that:

1

Improve
accuracy and
generalization

2

Handle complex
patterns

3

Scale efficiently
with data size

12



Motivation for Advanced Techniques

Motivation to move from:



Single models,
ensemble models



Simple clustering,
density- and model-
based clustering



Raw features, feature
engineering & dimensionality
reduction

13



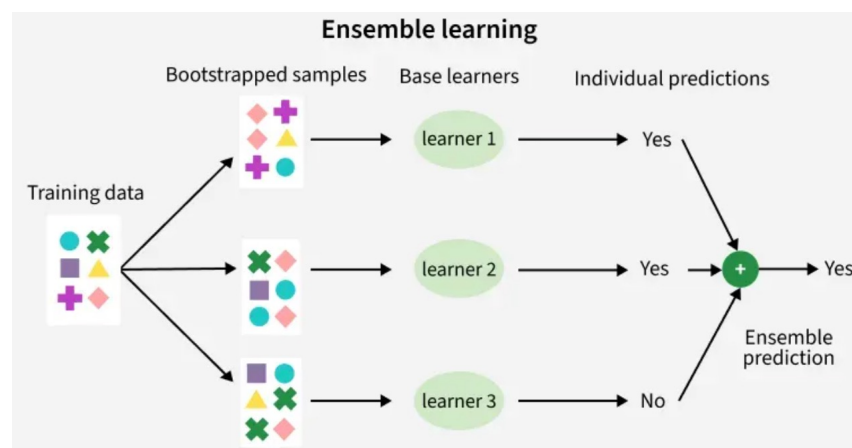
Transition to Part 2

Part 2 focuses on:

- Advanced machine learning techniques
- Scalability and efficiency
- Practical Big Data applications
- Builds a bridge from foundational concepts to real-world analytics



Advanced Supervised Learning





Advanced Supervised Learning

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Motivation

- Single models often struggle with:
 - Noisy data
 - Complex, non-linear patterns
 - Large-scale datasets
- Advanced methods aim to improve accuracy, robustness, and generalization.

16



Advanced Supervised Learning

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Ensemble Learning

- Combine multiple models to produce a stronger predictor.
- Main strategies:
 - Bagging: reduce variance (e.g., Random Forest)
 - Boosting: reduce bias (e.g., Gradient Boosting)
 - Stacking: combine diverse models
- Widely used in Big Data applications.

17



What is Ensemble Learning?

Ensemble Learning 	Several models 
A machine learning technique that combines multiple base models (weak learners) to build a more robust predictor.	Ensemble methods aggregate predictions from several models to improve performance.

18



Ensemble Learning overview

What is Ensemble Learning?

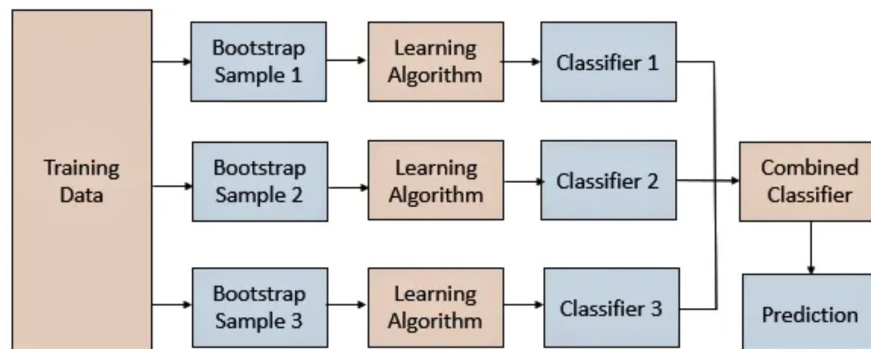
Big Data is often 	Ensemble models 
• Large-scale • Noisy • High-dimensional	• Improve prediction accuracy • Increase robustness • Reduce overfitting • Generalize better to unseen data

19



Main Ensemble Strategies

Bagging (Bootstrap Aggregating)



20



Main Ensemble Strategies

Key idea:

- Train multiple models independently on different bootstrap samples (random samples with replacement).
- Final prediction = average (regression) or majority vote (classification).

Main objective:

- Reduce variance

Example: Random Forest

- Many decision trees were trained on different data subsets
- Random feature selection increases diversity

21



Main Ensemble Strategies

Strengths:



Effective for unstable models (e.g., decision trees)



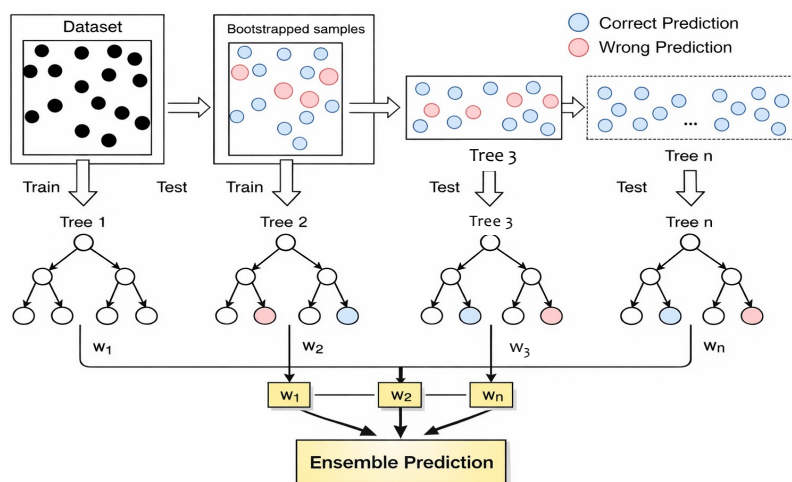
Easy to parallelize, suitable for Big Data based clustering

22



Main Ensemble Strategies

- Boosting



23



Main Ensemble Strategies

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Key idea:

- Models are trained sequentially.
- Each new model focuses on correcting errors made by previous models.

Main objective:

- Reduce bias (and sometimes variance)

Example:

- Gradient Boosting
- AdaBoost
- XGBoost (conceptual)

24



Main Ensemble Strategies

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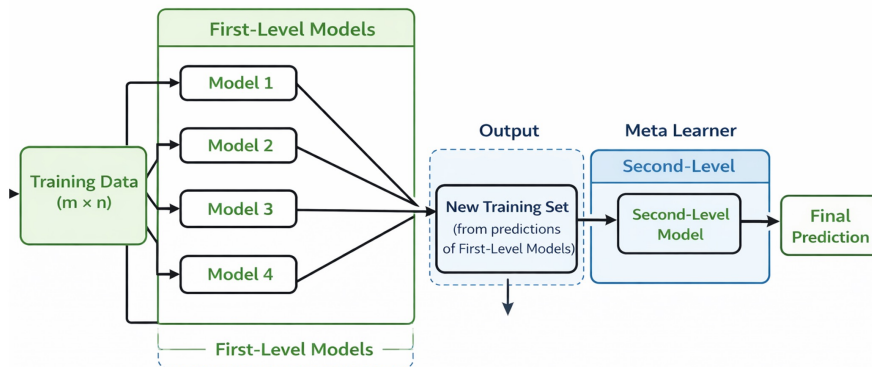
Strengths	Limitation
• High predictive accuracy • Handles complex, non-linear patterns well	• More sensitive to noise • Harder to parallelize than bagging

25



Main Ensemble Strategies

- Stacking (Stacked Generalization)



26



Main Ensemble Strategies

Key idea:

- Combine different types of models (e.g., decision tree, logistic regression, SVM).
- A meta-model learns how to best combine its predictions.

Workflow:

- Train base models
- Collect their predictions
- Train a meta-learner on these predictions

27



Main Ensemble Strategies

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Strengths	Limitation
• Leverages diversity among models • Often achieves strong performance in competitions	• More complex to implement • Higher computational cost

28



Comparison of Ensemble Strategies

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Strategy	Main Goal	Training Style	Typical Use Case
Bagging	Reduce variance	Parallel	Large, noisy data
Boosting	Reduce bias	Sequential	Complex patterns
Stacking	Combine strengths	Multi-level	High performance

29



Advanced Supervised Learning

Random Forest

- An ensemble of decision trees.
- Each tree is trained on:
 - A random subset of data
 - A random subset of features
- Advantages:
 - Handles high-dimensional data well
 - Robust to noise and overfitting
 - Easily parallelized for Big Data

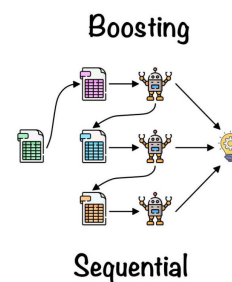
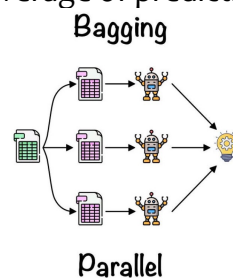
30



Random Forest-overview

What is Random Forest?

- Random Forest is an ensemble supervised learning algorithm.
- It builds many decision trees and combines their predictions.
- Final prediction:
 - Classification: majority voting
 - Regression: average of predictions



Source: Medium

31



How Random Forest Works?

Step-by-step Process

- From the original dataset, create multiple bootstrap samples (sampling with replacement).
- Train one decision tree per sample.
- At each split in a tree:
 - Randomly select a subset of features
 - Choose the best split only among those features.
- Aggregate predictions from all trees to produce the final output.
- **Key idea:** introduce randomness to increase model diversity.

32



Why “Random”?

Randomness is introduced in two ways	This helps
• Random sampling of data (bootstrapping) • Random sampling of features at each split	• Reduce correlation between trees • Reduce overfitting • Improve generalization

33



Bias–Variance Perspective

- Random Forest mainly reduces variance
- Very effective for noisy datasets

Model Type	Bias	Variance
Single Decision Tree	Low	High
Random Forest	Slightly higher	Much lower



Advantages of Random Forest

- Handles high-dimensional data well
- Robust to:
 - Noise
 - Outliers
- Works with:
 - Numerical data
 - Categorical data
- Easy to parallelize → suitable for Big Data
- Provides feature importance



Feature Importance

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- Measures how much each feature contributes to prediction
- Useful for:
 - Feature selection
 - Model interpretation
- Helps reduce dimensionality before applying complex models

36



Limitations of Random Forest

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- Large number of trees → high memory usage
- Less interpretable than a single decision tree
- Slower prediction time compared to simple models
- Not ideal for:
 - Very sparse data
 - Extrapolation problems

37



Random Forest in Big Data Context

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- Can be trained in parallel
- Used in:
 - Fraud detection
 - Customer churn prediction
 - Medical risk prediction
 - Recommendation systems
- Often a strong baseline model in real-world projects

38



Random Forest vs Decision Tree

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Aspect	Decision Tree	Random Forest
Overfitting	High	Low
Stability	Low	High
Accuracy	Moderate	High
Interpretability	High	Medium
Scalability	Limited	Good

39



Advanced Supervised Learning

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Gradient Boosting (Conceptual)

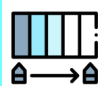
- Builds models sequentially.
- Each new model corrects errors of previous ones.
- Strengths:
 - High predictive performance
 - Effective for complex data patterns
- Common in finance, marketing, and risk prediction.

40



Gradient Boosting – Overview

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What is Gradient Boosting?

- Gradient Boosting is an ensemble supervised learning method.
- It builds models sequentially, where each new model focuses on correcting the errors made by previous models.
- Typically uses decision trees (shallow trees) as base learners.

41



Core Idea of Gradient Boosting

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Instead of training models independently, Gradient Boosting trains models step by step, learning from mistakes.

- Start with a simple initial model
- Measure its errors (residuals)
- Train the next model to predict those errors
- Add the new model to improve the overall prediction.

42



Conceptual Steps

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- Train an initial model (often a very simple one).
 - Compute the error (residual) between predictions and true values.
 - Train a new model to predict these residuals.
 - Add the new model to the ensemble with a learning rate.
 - Repeat the process until performance is satisfactory.
- Each model focuses on what the previous models did wrong.

43



Why “Gradient”?

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- Gradient Boosting uses gradient descent ideas.
- The model is improved by moving in the direction that minimizes the loss function.
- No need for SV to know derivatives here - just remember: Each new model reduces the overall error step by step.

44



Key Characteristics

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- Sequential learning (not parallel like Bagging)
- Strong ability to model:
 - Non-linear relationships
 - Complex feature interactions
- Highly flexible through:
 - Choice of loss function
 - Tree depth
 - Learning rate

45



Strengths of Gradient Boosting

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- Very high predictive performance
- Excellent for complex and structured data
- Works well with:
 - Tabular data
 - Mixed feature types
- Frequently wins ML competitions

46



Limitations of Gradient Boosting

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- Sensitive to:
 - Noise
 - Outliers
- Slower training due to sequential nature
- Requires careful hyperparameter tuning
- Risk of overfitting if not regularized

47



Gradient Boosting vs Random Forest

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Aspect	Random Forest	Gradient Boosting
Training	Parallel	Sequential
Main goal	Reduce variance	Reduce bias
Sensitivity to noise	Low	Higher
Accuracy	High	Very high
Tuning effort	Low	High

48



Popular Gradient Boosting Variants

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- Gradient Boosting Machines (GBM)
- XGBoost – optimized for speed & regularization
- LightGBM – efficient for large datasets
- CatBoost – strong with categorical features

49



Applications of Gradient Boosting

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- Finance:
 - Credit scoring
 - Fraud detection
- Marketing:
 - Customer churn prediction
 - Campaign response modeling
- Risk prediction:
 - Insurance
 - Healthcare

50



Advanced Unsupervised Learning

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- Why Advanced Unsupervised Learning?
- Real-world Big Data is often:
 - High-dimensional
 - Noisy
 - Non-spherical in structure
- Basic methods (e.g., K-means) have strong assumptions.
- Advanced methods relax these assumptions to discover more realistic patterns.

51



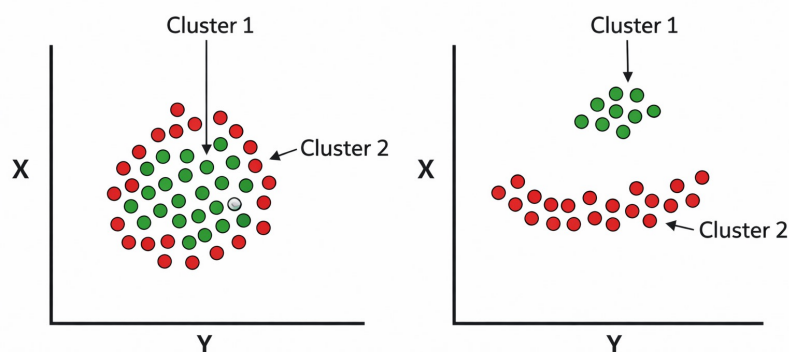
Limitations of Basic Clustering

- K-means:
 - Assumes spherical clusters
 - Requires predefined number of clusters (K)
 - Sensitive to outliers
- Hierarchical clustering:
 - High computational cost
 - Hard to scale to very large datasets
- Motivation to move toward density-based and model-based approaches.



Density-Based Clustering

DBScan Clustering





Density-Based Clustering

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Key Idea

- Clusters are defined as regions of high data density.
- Points in low-density regions are treated as noise.

Characteristics

- Can find clusters of arbitrary shapes
- No need to predefine number of clusters
- Robust to noise and outliers

54



DBSCAN (Conceptual)

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Groups points based on:

- Neighborhood radius (ϵ)
- Minimum number of points

Classifies points into:

- Core points
- Border points
- Noise

55



DBSCAN (Conceptual)

Strengths

- Works well with irregular cluster shapes
- Handles noise explicitly

Limitations

- Sensitive to parameter selection
- Struggles with varying densities



HDBSCAN (Conceptual)

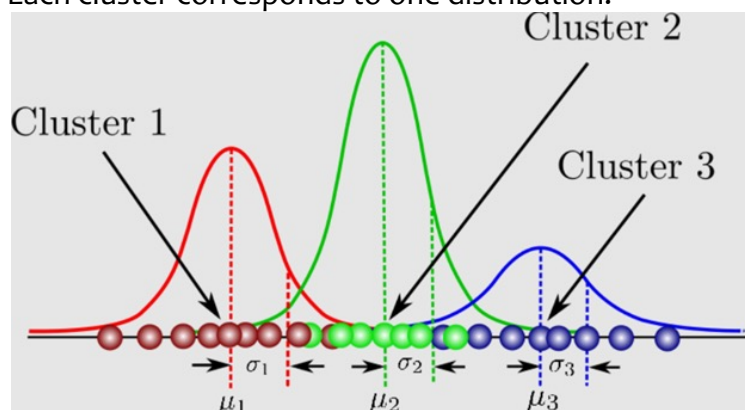
- Hierarchical extension of DBSCAN
- Automatically determines:
 - Number of clusters
 - Cluster stability
- **Better suited for Big Data with varying densities**



Model-Based Clustering

Key Idea

- Data is assumed to be generated by a mixture of probability distributions.
- Each cluster corresponds to one distribution.



Source: Medium

58



Gaussian Mixture Models (GMM)

- Uses multiple Gaussian distributions
- Each data point has a probability of belonging to each cluster.
- Allows:
 - Soft clustering (not hard assignment)
 - Elliptical cluster shapes

59


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Gaussian Mixture Models (GMM)

Advantages

- More flexible than K-means
- Provides probabilistic interpretation

Limitations

- Assumes underlying distribution
- Sensitive to initialization
- Computationally expensive for large datasets

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60


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Comparison of Advanced Unsupervised Methods

Method	Cluster Shape	Handles Noise	Scalability	Notes
K-means	Spherical	No	High	Simple, fast
DBSCAN	Arbitrary	Yes	Medium	Parameter sensitive
HDBSCAN	Arbitrary	Yes	Better	Automatic clustering
GMM	Elliptical	Limited	Medium	Probabilistic

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61

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Advanced Unsupervised Learning in Big Data

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Customer segmentation



Anomaly detection



Image & signal clustering



Bioinformatics



Network & graph analysis
(conceptual link)

62

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Key Takeaways

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- Advanced unsupervised methods handle complex and noisy data.
- Density-based methods focus on data distribution in space.
- Model-based methods provide probabilistic cluster membership.
- Method choice depends on:
 - Data shape
 - Noise level
 - Scalability needs

63

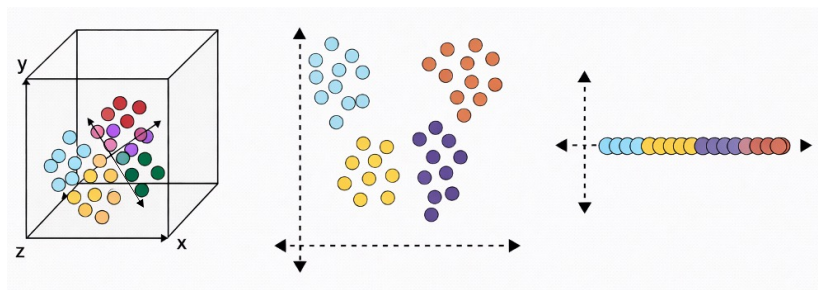


Dimensionality Reduction



What is Dimensionality Reduction?

- Dimensionality Reduction is the process of reducing the number of features (dimensions) in a dataset.
- The goal is to retain the most important information while removing redundancy and noise.





Why Dimensionality Reduction in Big Data?

Key Challenges

- Big Data often has:
 - Thousands of features
 - Sparse and noisy attributes
- High dimensionality leads to:
 - Increased computation cost
 - Poor model performance
 - Difficulty in visualization
- This phenomenon is known as the curse of dimensionality.



Benefits of Dimensionality Reduction



Improves model performance



Reduces overfitting



Lowers storage and computation cost



Enables data visualization (2D/ 3D)



Simplifies downstream machine learning tasks



Main Approaches

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1. Feature Selection

- Select a subset of original features.
- Remove irrelevant or redundant features.
- Original feature meaning is preserved.

Examples:

- Correlation-based selection
- Tree-based feature importance

68



Main Approaches

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2. Feature Extraction

- Transform original features into a new feature space.
- New features are combinations of original ones.
- Original feature meaning may be lost.

Examples:

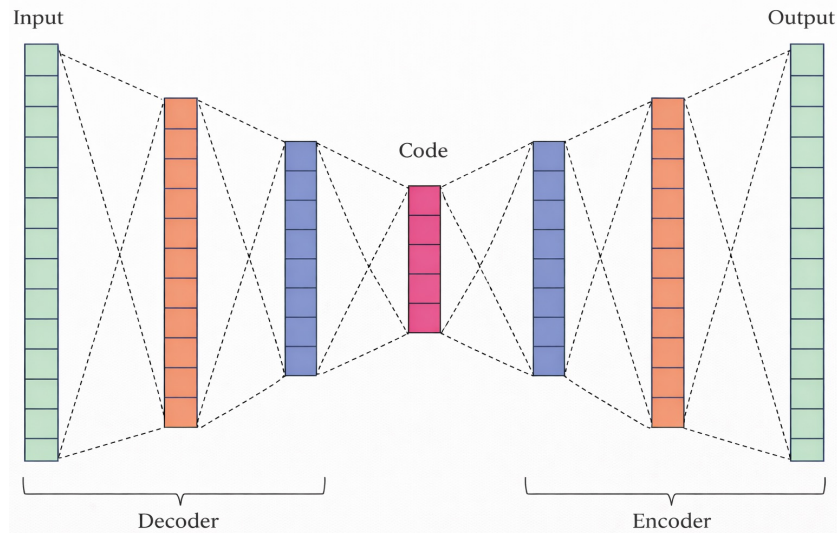
- PCA
- Autoencoders

69



Common Dimensionality Reduction Techniques

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70



Common Dimensionality Reduction Techniques

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PCA (Principal Component Analysis)

- Linear technique
- Projects data onto directions of maximum variance
- Widely used for preprocessing and visualization

t-SNE

- Non-linear technique
- Preserves local structure
- Mainly used for visualization (not modeling)

71



Common Dimensionality Reduction Techniques

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UMAP

- Non-linear and scalable
- Preserves both local and global structure
- Suitable for large datasets

Autoencoders

- Neural network-based approach
- Learn compressed representations
- Effective for very high-dimensional data

72



Dimensionality Reduction in Big Data Pipelines

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- Applied before clustering or classification
- Used to:
 - Speed up training
 - Improve clustering quality
- Often combined with:
 - Feature engineering
 - Ensemble learning

73

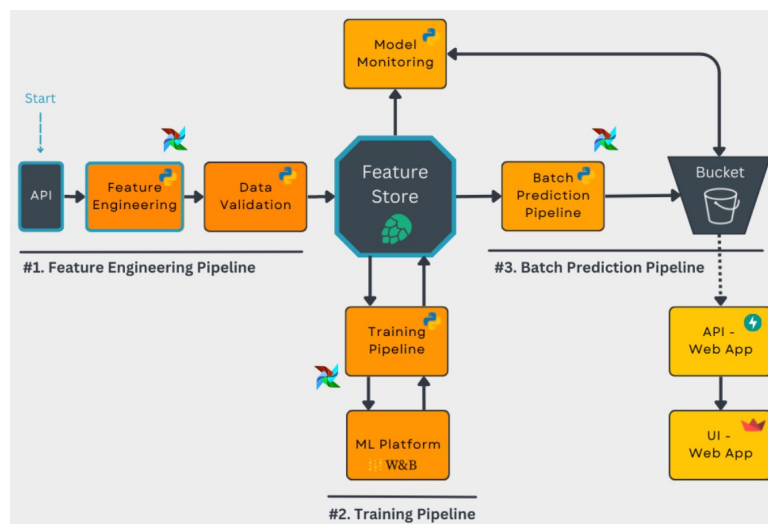


Key Takeaways

- Dimensionality reduction addresses challenges of high-dimensional Big Data.
- Improves efficiency, performance, and interpretability.
- Can be achieved via feature selection or feature extraction.
- Choice of method depends on:
 - Data size
 - Modeling goal
 - Visualization needs



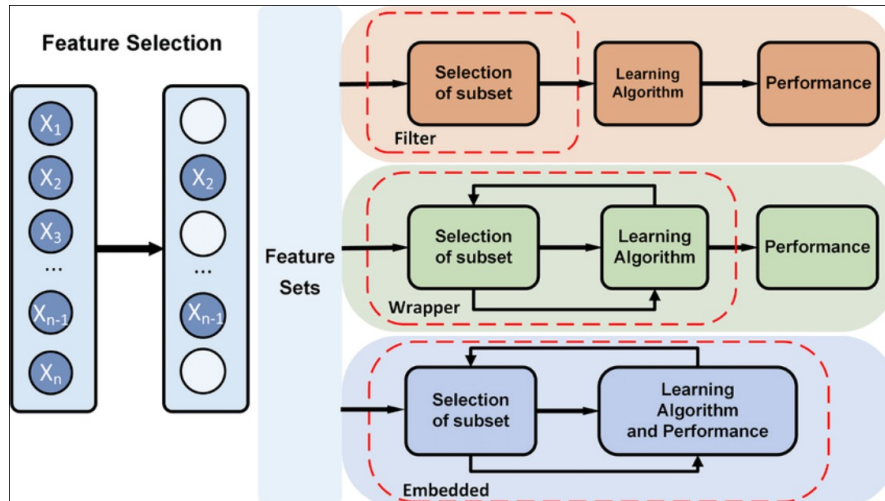
Feature Engineering & Feature Selection





Feature Engineering & Feature Selection

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Source: blog.devgenius

76



Why Features Matter

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- In many ML problems, features are more important than the choice of model.
- Poor features → poor performance, even with advanced algorithms.
- Big Data often contains:
 - Redundant features
 - Noisy features
 - Irrelevant attributes

77



Feature Engineering

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What is Feature Engineering?

- The process of creating, transforming, or improving features to better represent the underlying problem.
- Relies heavily on:
 - Domain knowledge
 - Data understanding

78



Common Feature Engineering Techniques

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- Feature construction
 - Combine existing features (e.g., ratios, differences)
- Feature transformation
 - Normalization / standardization
 - Log or power transformation
- Encoding
 - One-hot encoding
 - Label encoding
- Aggregation
 - Mean, count, max, min over time or groups

79



Feature Engineering in Big Data

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- Helps models:
 - Capture meaningful patterns
 - Handle heterogeneous data sources
- Often applied in:
 - Customer analytics
 - Transaction data
 - Time-series and log data

80



Feature Selection

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What is Feature Selection?

- The process of selecting a subset of relevant features from the original feature set.
- Goal:
 - Remove irrelevant or redundant features
 - Improve efficiency and generalization

81



Benefits of Feature Selection

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- Reduces dimensionality
- Speeds up model training
- Improves interpretability
- Reduces overfitting

82



Main Feature Selection Methods

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1. Filter Methods

- Based on statistical properties of data
- Independent of any ML model
- Examples:
 - Correlation
 - Variance threshold
- Fast, scalable
- Ignore model interaction

83



Main Feature Selection Methods

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2. Wrapper Methods

- Use a predictive model to evaluate feature subsets
- Examples:
 - Forward selection
 - Backward elimination
- Better performance
- Computationally expensive

84



Main Feature Selection Methods

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3. Embedded Methods

- Feature selection occurs during model training
- Examples:
 - Tree-based feature importance
 - Regularization-based methods
- Balanced efficiency and performance
- Popular in Big Data workflows

85



Feature Engineering vs Feature Selection

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Aspect	Feature Engineering	Feature Selection
Purpose	Create better features	Remove unnecessary features
Output	New feature space	Reduced feature set
Domain knowledge	Very important	Less critical
Use in pipeline	Early stage	After feature creation

86



Role in Big Data Analytics

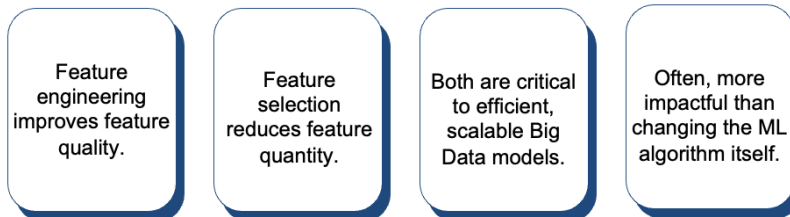
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- Typically combined with:
 - Dimensionality reduction
 - Ensemble learning
- Essential for:
 - Scalability
 - Model robustness
- Strong impact on real-world ML systems

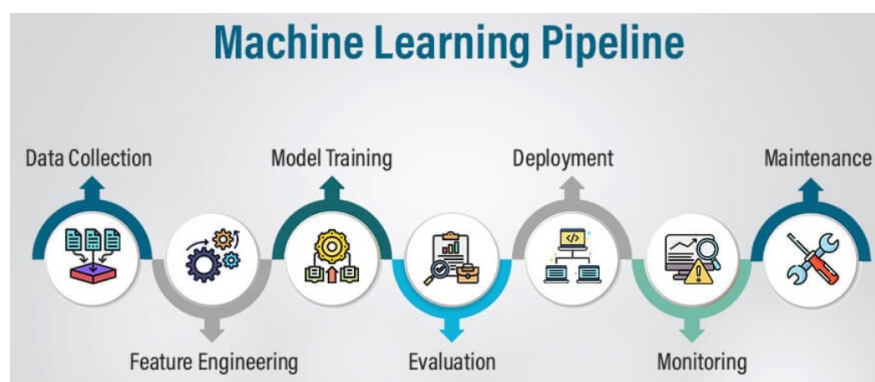
87



Key Takeaways



Machine Learning at Scale





What is Machine Learning at Scale?

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- Machine Learning at Scale refers to applying ML techniques on:
 - Very large datasets
 - High-dimensional data
 - Distributed computing environments
- Focuses on scalability, efficiency, and robustness, not just accuracy.

90



Why Scalability Matters in Big Data?

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- Big Data is characterized by:
 - Large volume
 - High velocity
 - High variety
- Traditional ML algorithms often:
 - Do not fit into memory
 - Are too slow on a single machine
- Need scalable learning strategies.

91



Key Challenges

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- Data volume
 - Data cannot fit into a single machine's memory
- Computation cost
 - Training time increases rapidly
- Communication overhead
 - Data and model synchronization across nodes
- Model complexity
 - Large models require more resources

92



Core Concepts of Machine Learning at Scale

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1. Distributed Data Processing

Data is split
across multiple
machines
(nodes)

Each node
processes a
subset of data

Results are
aggregated
centrally

93



Core Concepts of Machine Learning at Scale

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2. Parallel Model Training

- Train multiple models or model components simultaneously
- Common in:
 - Ensemble methods
 - Tree-based models
- Reduces training time significantly

94



Core Concepts of Machine Learning at Scale

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3. Batch vs Streaming Learning

Aspect	Batch Learning	Streaming Learning
Data	Static datasets	Continuous data
Training	Periodic	Incremental
Use cases	Offline analytics	Real-time systems

95



Core Concepts of Machine Learning at Scale

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4. Model Aggregation

- Combine partial results from different nodes
- Examples:
 - Averaging model parameters
 - Voting / ensemble aggregation

96



Scalable ML Frameworks (Conceptual)

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- Distributed ML libraries
- In-memory data processing engines
- Support:
 - Parallelism
 - Fault tolerance
 - Scalability

97



Machine Learning at Scale in Practice

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Fraud detection systems



Recommendation engines



Customer behavior analysis



Healthcare risk prediction



Smart city analytics

98



Key Takeaways

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- Machine Learning at Scale focuses on handling large and complex data efficiently.
- Scalability is as important as model accuracy.
- Parallelism and distributed processing are key enablers.
- Essential for real-world Big Data applications.

99



Applications & Case Studies

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Application 1: Fraud Detection (Finance)

- Context: A fraud detection system must process millions of transactions daily. False positives increase operational cost.

Application Tasks:

- You must prioritize robustness OR interpretability, not both. Which one do you choose, and why?
- Which technique from Lec. 5 becomes less suitable under this constraint?
- If data noise increases, which technique degrades least in performance?

Target concepts: Random Forest, Gradient Boosting, Ensemble robustness

100



Applications & Case Studies

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Application 2: Customer Analytics (Marketing & CRM)

- Context: A company wants to understand customer behavior, not just predict outcomes.

Application Tasks:

- Why is prediction accuracy alone insufficient in this application?
- Which technique from Lec. 5 helps discover structure, not just predict?
- Under what condition would ensemble learning be overkill here?

Target concepts: Advanced Unsupervised Learning, Clustering vs Classification

101


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Applications & Case Studies

Application 3: Healthcare Analytics

- Context: A hospital uses ML to support clinical decision-making. Doctors need to trust the system.

Application Tasks

- Which technique from Lec. 5 creates the biggest trust issue?
- How can feature selection reduce (but not eliminate) this issue?
- In this application, when is slightly lower accuracy acceptable?

Target concepts: Accuracy vs Interpretability, Feature Selection, Ensemble learning

102


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Applications & Case Studies

Case 1: Fraud Detection in Online Banking

- Context: A bank processes millions of transactions daily. Fraud cases are rare, data is noisy, and false alarms are expensive.

Discussion Tasks

- Choose ONLY ONE technique from Lec. 5 and justify your choice by:
 - (1) describing the data characteristics,
 - (2) explaining why this technique fits those characteristics,
 - (3) identifying at least one trade-off of your choice.
- Explain why at least TWO other techniques from Lec. 5 are NOT suitable for this case.
- What trade-off does your chosen technique introduce?
- If transaction volume doubles, what part of your approach breaks first?

Focus concepts: Ensemble Learning, Random Forest vs Gradient Boosting, ML at scale

103


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Applications & Case Studies

Case 2: Smart City Traffic Monitoring

Context: Traffic sensors continuously stream data. The city wants congestion patterns with minimal delay.

Discussion Tasks

- Batch learning is currently used. Identify ONE specific limitation of batch learning in this scenario, based on data velocity or latency requirements.
- Choose ONE concept from Lec. 5 that directly addresses this limitation, and explain how it helps.
- If real-time constraints are strict, which technique from Lec. 5 should be avoided? Why?

Focus concepts: Machine Learning at Scale, Streaming vs Batch, Scalability

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104


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Applications & Case Studies

Case 3: Healthcare Risk Prediction

Context: A hospital wants to predict patient risk. Accuracy matters, but doctors require explanations.

Discussion Tasks

- You may choose ONLY ONE model family from Lec. 5. Briefly describe which requirement you prioritize (accuracy or interpretability) and why this priority is reasonable in this context.
- Identify ONE interpretability issue introduced by your chosen model family, and explain why this issue is problematic for clinicians.
- How can feature selection partially mitigate this issue?
- Under what condition would you reject ensemble learning entirely for this application?

Focus concepts: Ensemble Learning · Feature Selection · Accuracy vs Interpretability

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105



Thank you for your attention!